EMPLOYEE ATTRITION & WORKPLACE ANALYTICS

Exploratory Data Analysis Report

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Dataset: IBM HR Analytics Employee Attrition & Performance

Records: 1,470 in the standard dataset

Project Purpose

This report investigates employee attrition patterns and compares employees who left with employees who stayed. It answers the ten assignment questions using descriptive statistics and group comparisons.

Note: Figures in this report are rounded approximations for the commonly distributed 1,470-row dataset. Verify them against the exact CSV used for submission.

1. Dataset and Methodology

The dataset records employee attributes such as age, department, job role, monthly income, overtime status, satisfaction ratings, distance from home, tenure, and an Attrition label (Yes/No).

Methods include data inspection, missing-value and duplicate checks, descriptive statistics, group-by attrition rates, bar plots and box plots, a numerical correlation heatmap, IQR-based outlier screening, and optional statistical tests.

2. Answers to Assignment Questions

Q1. What is the overall attrition rate?

237 of 1,470 employees are marked as having left, while 1,233 stayed. The attrition rate is 237 / 1,470 × 100 ≈ **16.12%**; retention is approximately 83.88%.

Q2. Which departments have higher attrition?

Sales has an attrition rate of approximately **20.63%**, Human Resources about **19.05%**, and Research & Development about **13.84%**. Rates are more useful than raw departure counts because department sizes differ.

Q3. Does overtime relate to attrition?

The observed attrition rate is approximately **30.53%** for employees working overtime versus **10.44%** for employees not working overtime. This is an association in the dataset, not proof that overtime causes attrition.

Q4. Does income differ between employees who leave and stay?

Mean monthly income is approximately **4,787** among employees who left and **6,833** among those who stayed. Income also varies with job role, experience, and seniority, so this difference should not be interpreted in isolation.

Q5. How does job satisfaction relate to attrition?

The attrition rate is about **21.45%** for satisfaction score 1 and **11.55%** for score 4. This suggests lower satisfaction is associated with higher attrition in this dataset, although individual outcomes vary.

Q6. Does distance from home relate to attrition?

Employees who left have a mean distance-from-home value of about **10.63**, compared with about **8.92** for employees who stayed. This group difference does not prove that commuting distance caused attrition.

Q7. Which job roles have higher attrition?

Sales Representatives have the highest role-level attrition rate, approximately **39.76%** (33 departures among 83 employees). Laboratory Technicians are also comparatively high at about 23.94%. Consider group sizes when interpreting these rates.

Q8. Does age influence attrition patterns?

Employees who left are younger on average (about **33.61 years**) than employees who stayed (about **37.56 years**). This is an observed pattern, not evidence that age alone causes attrition.

Q9. How does work-life balance relate to retention?

The lowest work-life balance score (1) has an attrition rate of about **31.25%**. Scores 2 and 3 are about 16.86% and 14.22%, while score 4 is about 17.65%. The pattern is not perfectly monotonic and should be interpreted cautiously.

Q10. What differences are observed between employees who left and stayed?

In the standard dataset, leavers are younger on average, have lower mean monthly income, and have a higher average distance-from-home value. The notebook's leavers-versus-stayers table compares these and other variables, including tenure, job satisfaction, and work-life balance.

3. Summary of Key Results

Measure	Observed result
Overall attrition	237 / 1,470; approximately 16.12%
Highest department rate	Sales; approximately 20.63%
Overtime comparison	Yes: 30.53%; No: 10.44%
Mean monthly income	Left: ~4,787; Stayed: ~6,833
Job satisfaction	Score 1: ~21.45%; score 4: ~11.55%
Mean distance from home	Left: ~10.63; Stayed: ~8.92
Highest job-role rate	Sales Representative; ~39.76%
Mean age	Left: ~33.61; Stayed: ~37.56
Work-life balance	Score 1: ~31.25%; score 4: ~17.65%

4. Business Recommendations

• Investigate departments and roles with higher attrition, particularly Sales and Sales Representatives.

• Review overtime practices, staffing levels, workload, and recovery time.

• Examine compensation by role, experience, and seniority rather than relying only on overall averages.

• Monitor job satisfaction and work-life balance and follow up where ratings are low.

• Explore career development and retention support for younger employees and early-career roles.

• Validate potential interventions with employee feedback and further analysis.

5. Conclusion

The EDA indicates that attrition is associated with several employee and workplace characteristics. Overtime, job role, income, satisfaction, age, and work-life balance are useful areas for further investigation. These descriptive results can guide follow-up analysis but do not prove that any single factor causes employees to leave.

6. Limitations and Submission Note

This is observational data from one dataset. Category sizes differ, group averages can hide variation, and unmeasured factors may influence outcomes. Correlation and statistical significance do not establish causation.

Before submitting: run the notebook from top to bottom using the exact CSV you include. Check the generated summaries and update figures if your CSV's results differ from the standard dataset.

Dataset source: Kaggle — IBM HR Analytics Employee Attrition & Performance:
https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset